Assignment-15

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Batch: 37

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Lab: 15.2

Task Descriptor-1: (Setup Fast API Backend)
Instructions:

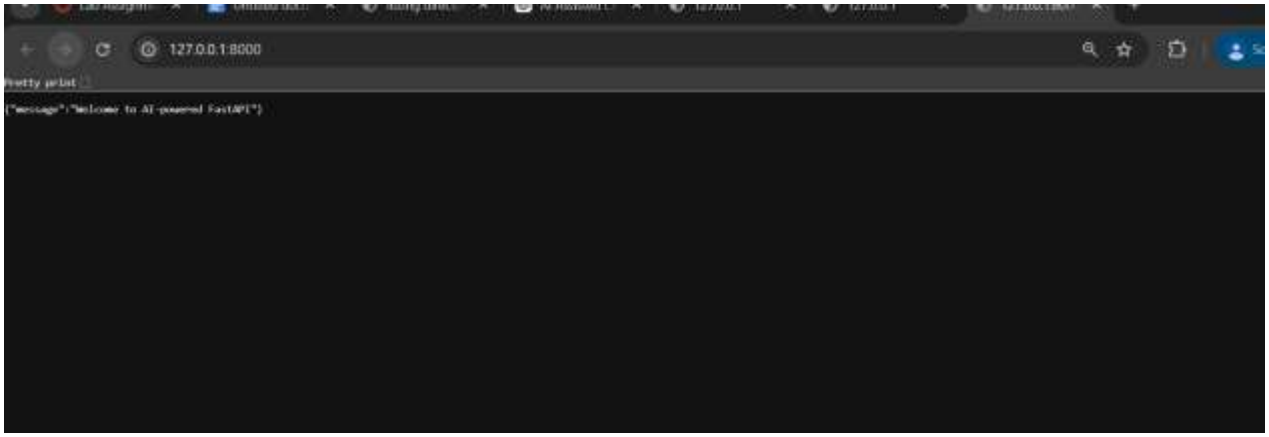
- Install Fas tAPI and Uvicorn.
- Use AI to generate a basic FastAPI application with one endpoint
  Sample Code: from fastapi import FastAPI app = FastAPI()
  @app.get("/") def read_root():

return {"message": "Welcome to AI-powered FastAPI"}
Expected Output:
Running the server and accessing:
http://127.0.0.1:8000/
Response:
{"message": "Welcome to AI-powered FastAPI"}

Prompt:-
Generate a simple FastAPI application in Python.
Requirements:
- Import FastAPI
- Create FastAPI app
- Create one GET endpoint "/"
- The endpoint should return JSON message:
  {"message": "Welcome to AI-powered FastAPI"}

The code should run using uvicorn and be beginner friendly.

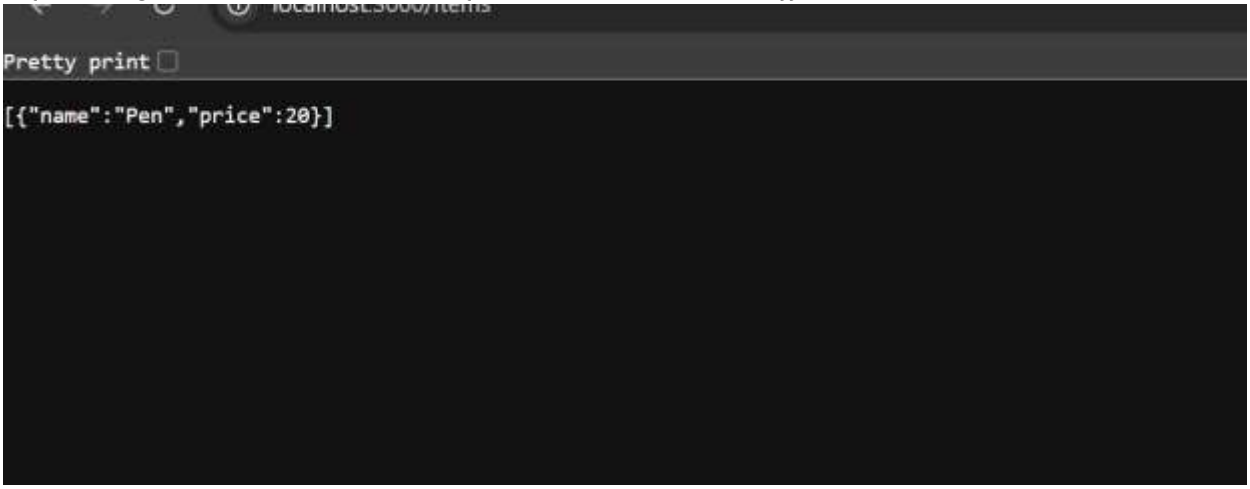
Output:-



Task Descriptor- 2: (API – Create and Read Items) Instructions:

- Implement endpoints to add and retrieve products.
- Use a Python list as temporary storage.

Sample Code: items = []
@app.get("/items") def
get_items(): return
items
@app.post("/items") def
add_item(item: dict):
items.append(item)
return {"message": "Item created", "item": item} Expected
Output:
GET /items → []
POST /items with {"name":"Pen","price":20}
→ {"message":"Item created","item":{"name":"Pen","price":20}}

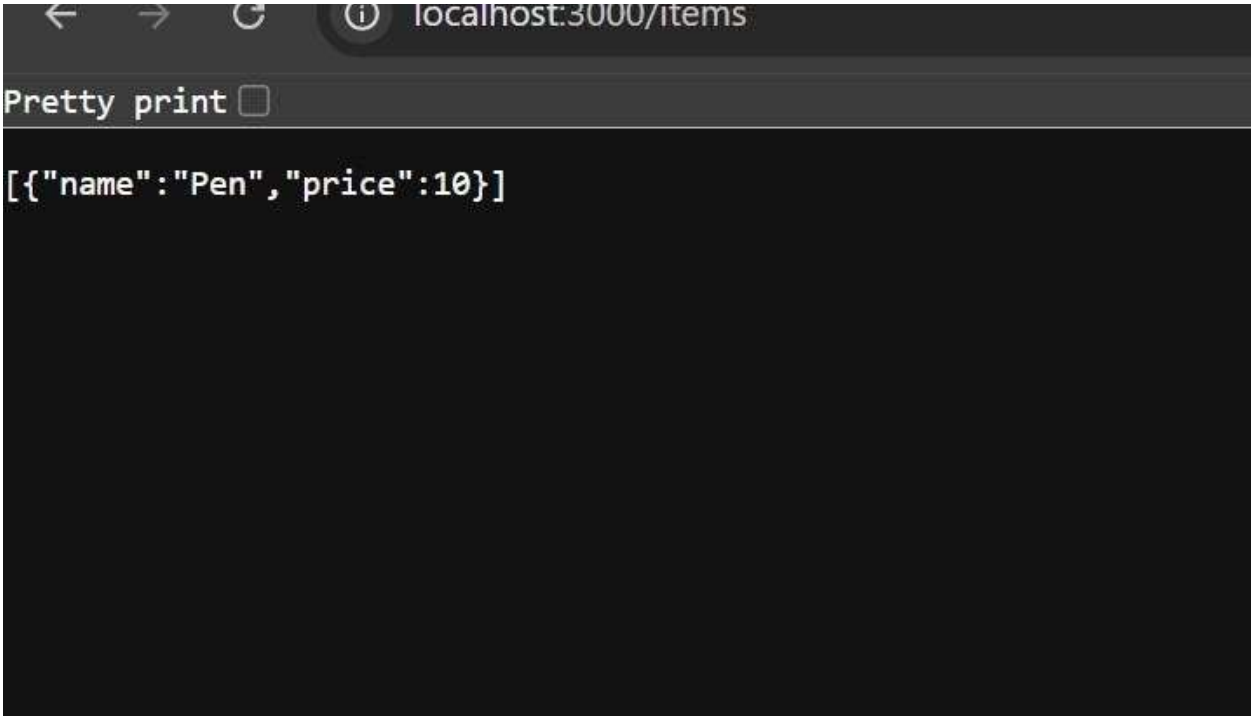


Task Descriptor- 3: (Update Existing Item) Instructions:

- Use AI to generate an endpoint to update an item using its index.

Sample Code:

```
@app.put("/items/{index}")          def
update_item(index: int, item: dict): if
index >= len(items): return {"error":
"Item not found"} items[index] = item
return {"message": "Item updated", "item": item} Expected
Output:
PUT /items/0 with {"name":"Pencil","price":10}
→ {"message":"Item updated","item":{"name":"Pencil","price":10}}
```

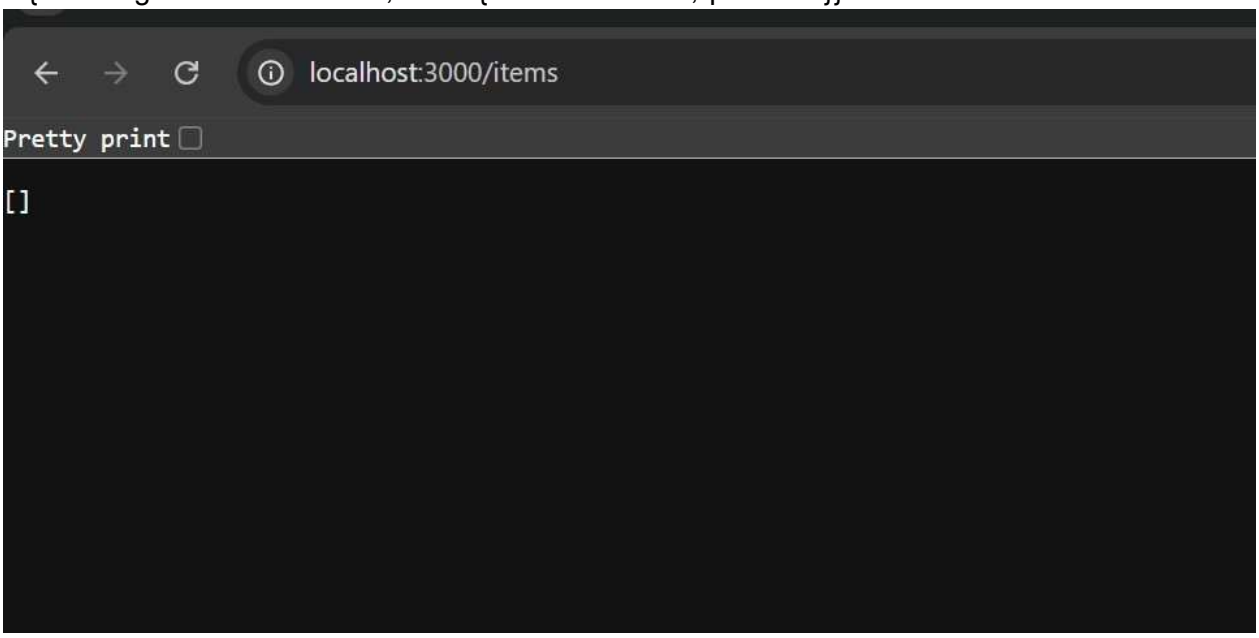


Task Descriptor- 4: (Delete Item) Instructions:

- Implement deletion of an item by index. Sample Code: @app.delete("/items/{index}") def delete_item(index: int): if index >= len(items):

```
return {"error": "Item not found"} removed
= items.pop(index)
return {"message": "Item removed", "item": removed} Expected
Output:
```

DELETE /items/0
→ {"message":"Item removed","item":{"name":"Pencil","price":10}}



Task Descriptor -5: (Add Auto-Generated Documentation)
Instructions:

- Use AI to add inline comments and docstrings for all endpoints.
- Optionally, integrate Swagger / Flask-RESTX to auto-generate API documentation.

Sample Code:

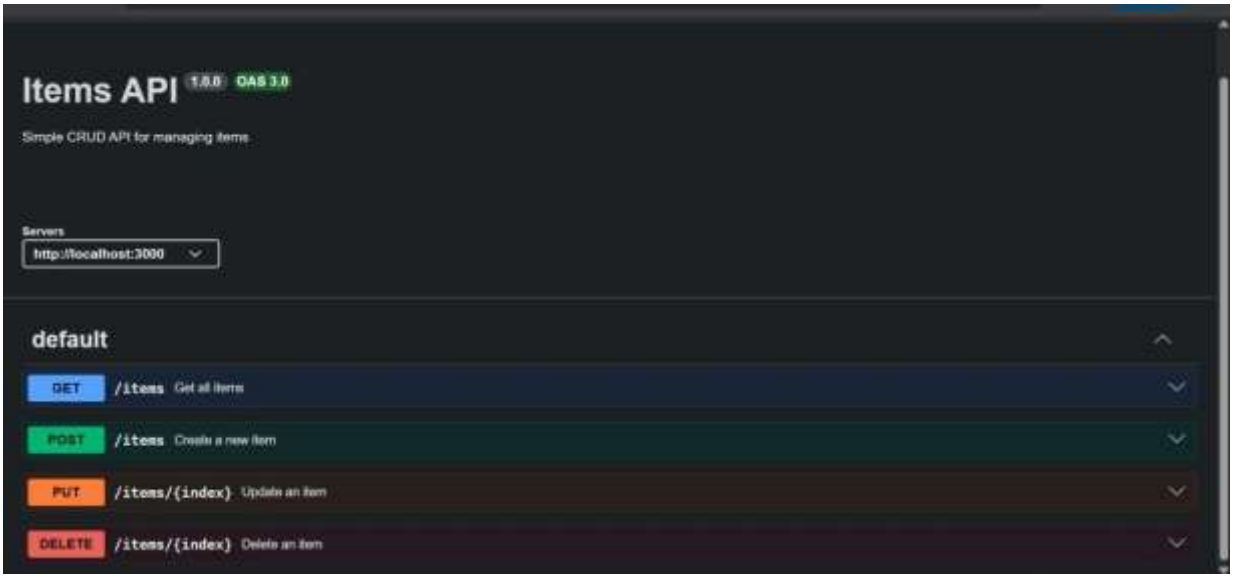
```
@app.route('/items', methods=['GET'])
def get_items(): """

GET /items
Returns a list of all items in the store.
"""
return jsonify(items)
```

Expected Output:

- Clear documentation for all endpoints, e.g., via /docs if using Swagger.

- Students should be able to see endpoint description, methods, and sample payloads.


Items API
1.0.0
OAS 3.0
Simple CRUD API for managing items
Servers
http://localhost:3000
default
GET
/items
Get all items
POST
/items
Create a new item
PUT
/items/{index}
Update an item
DELETE
/items/{index}
Delete an item